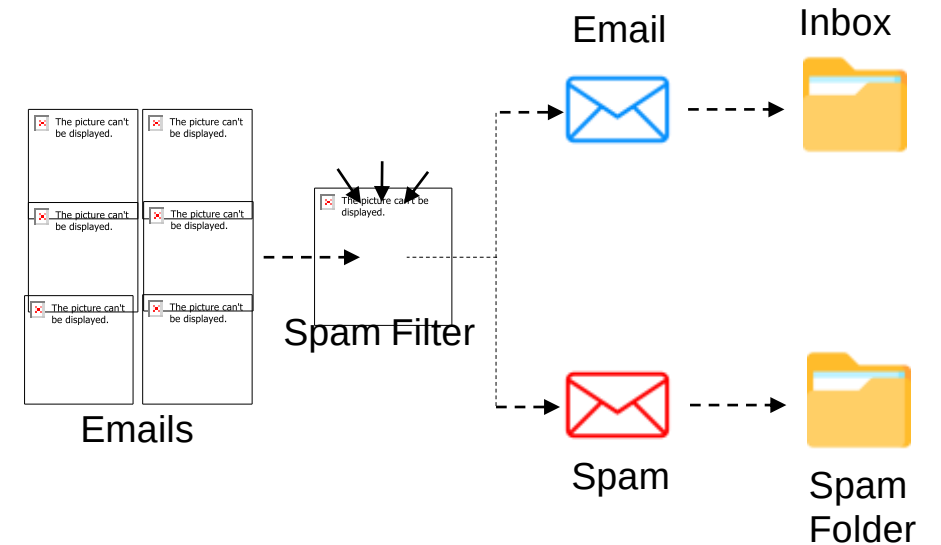


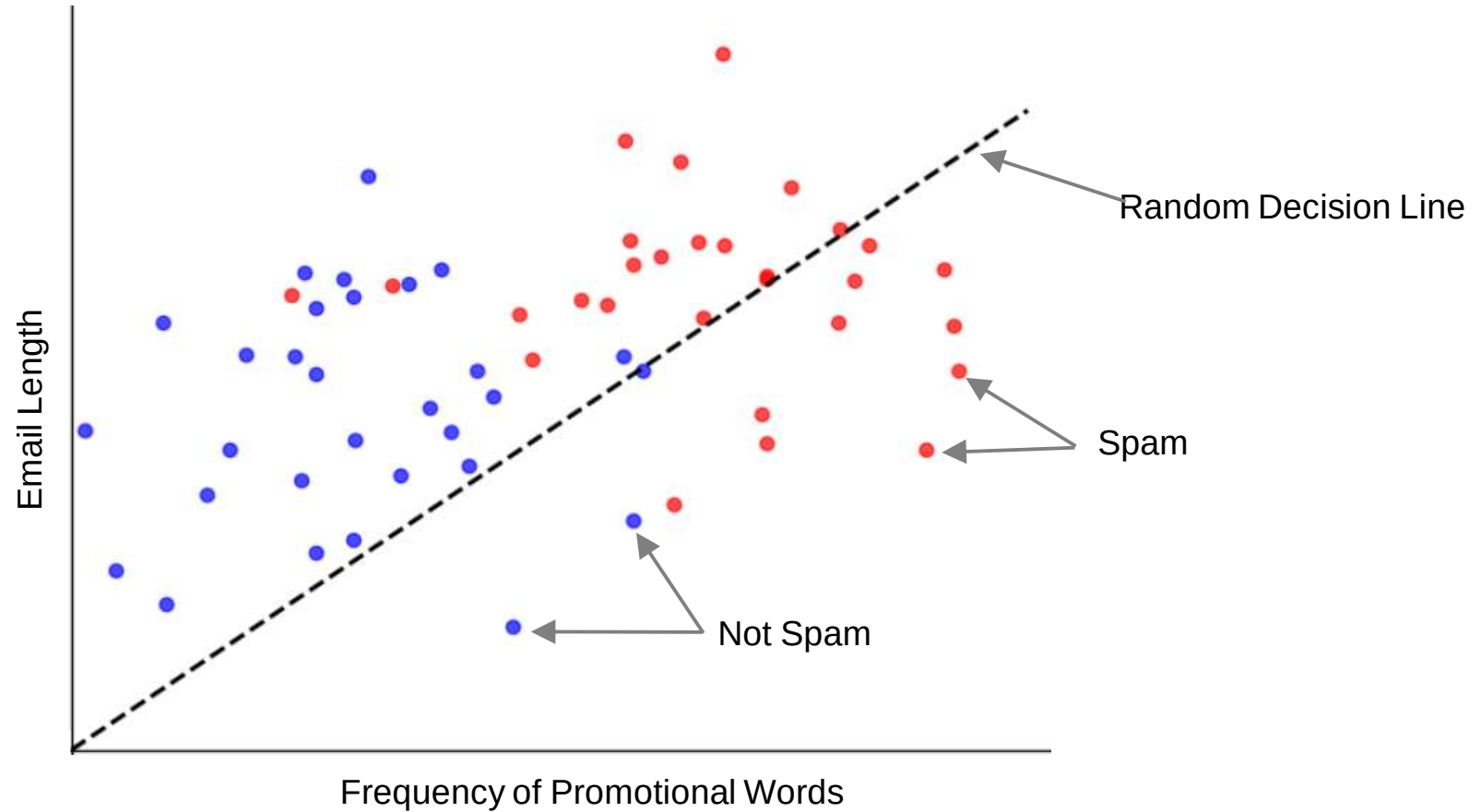
Need for a Probabilistic Model

NEED FOR A PROBABILISTIC MODEL

- Suppose you are building an email filter that decides whether a message is spam or not
- Some words (such as free, offer and lottery) increase suspicion, but they also appear in genuine mails
- A few messages are clearly spam, and a few clearly are not, but many lie in the grey area
- Is it possible to predict whether a mail is spam?



EMAIL CLASSIFICATION

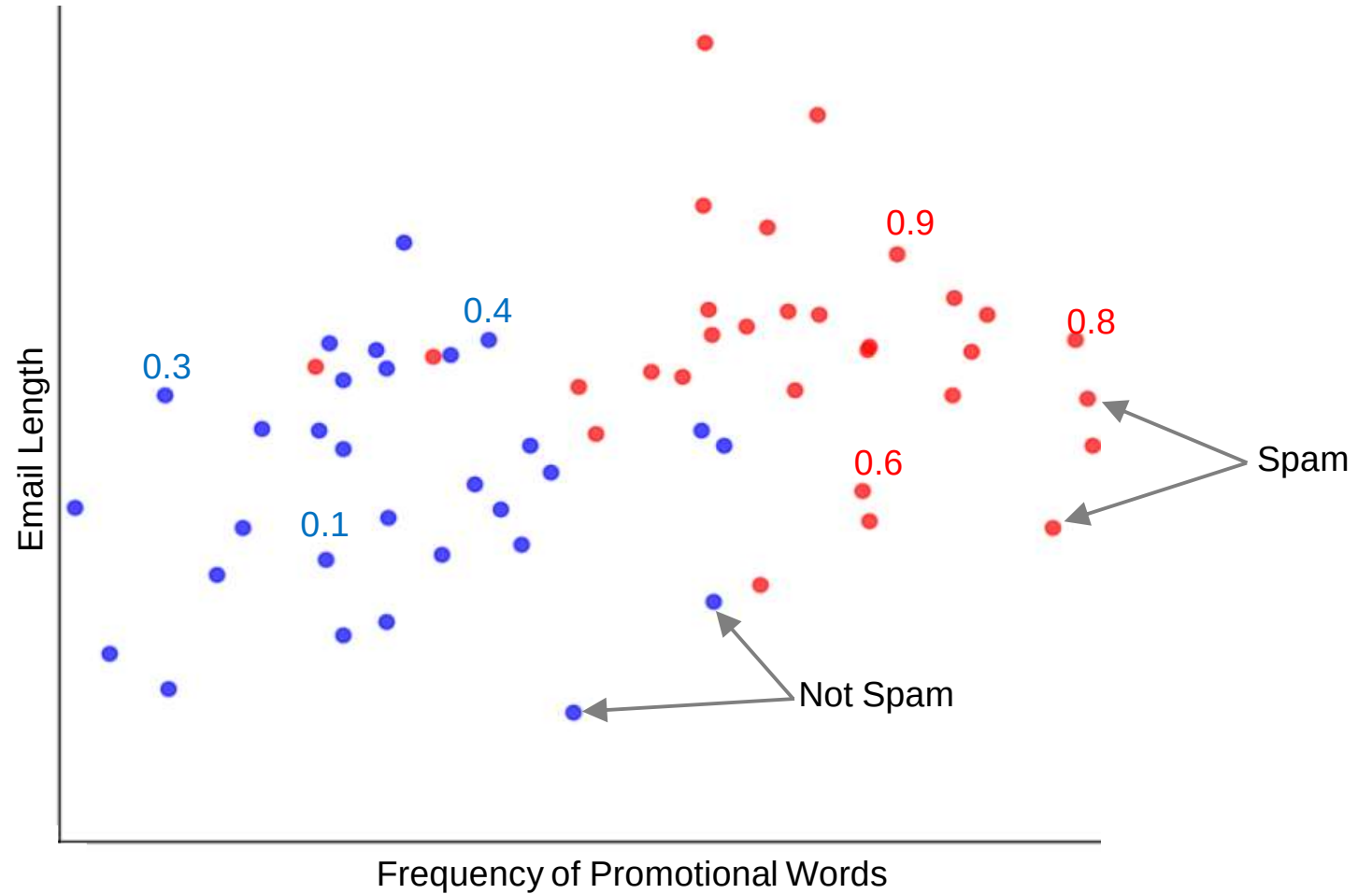


Real-world data rarely separates cleanly

Deterministic rules fail when classes overlap; therefore, we need a framework that accommodates *partial belief*

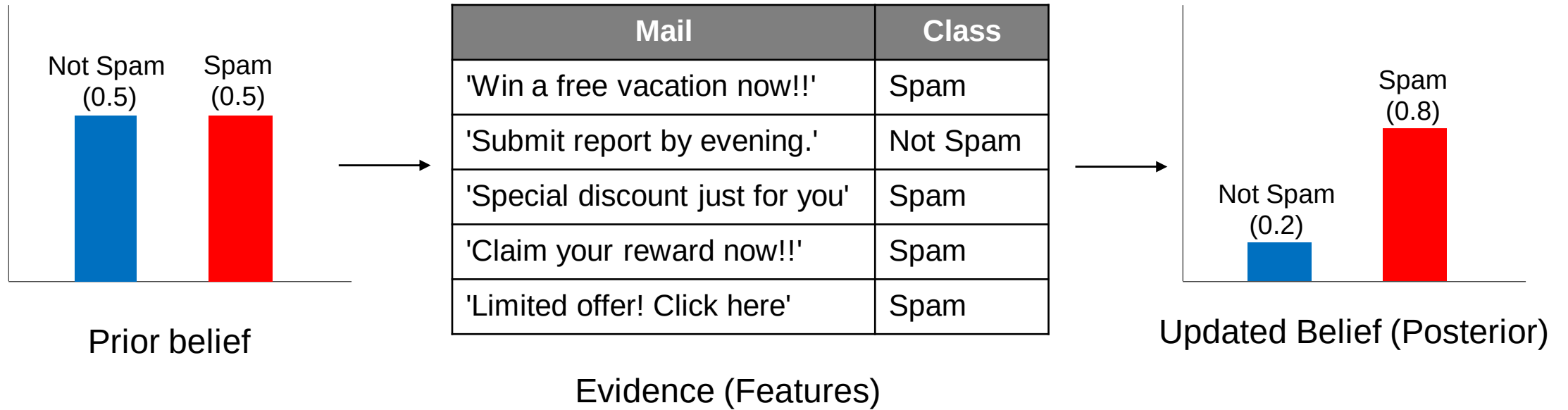
PROBABILISTIC CONFIDENCE IN EMAIL CLASSIFICATION

- Colours show the predicted class; numbers show the probability of the message being spam
- Confidence gradually changes across points, not across a sharp boundary
- A higher probability → Stronger belief



Probabilities capture *how sure* the model is, not just *what* it predicts

SOURCES OF PROBABILITIES



Probabilities emerge from how **evidence (features)** reshapes our **prior understanding** of each class

Conditional Probability

CONDITIONAL PROBABILITY

Conditional probability measures the probability of an event occurring, given that another event has already occurred (by assumption, presumption, assertion, or evidence)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Here:

- $P(A|B) \rightarrow$ Probability of event A occurring given that event B has already occurred
- $P(A \cap B) \rightarrow$ Probability that both events A and B occur together
- $P(B) \rightarrow$ Probability that event B occurs

EXAMPLE

Calculate the probability that a person has symptoms given that the person has already contracted COVID-19

	Contracted COVID-19	Not Contracted COVID-19	Total
Symptomatic	200	200	400
Asymptomatic	100	500	600
Total	300	700	1,000

$$\bullet P(A \cap B) = P(\text{Symptomatic} \cap \text{Contracted COVID} - 19) = \frac{200}{1,000} = 0.2$$

$$\bullet P(B) = P(\text{Contracted COVID} - 19) = \frac{300}{1,000} = 0.3$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{0.2}{0.3} = 0.67$$

Bayes Theorem

BAYES THEOREM

- We know the following conditional probabilities

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{and} \quad P(B|A) = \frac{P(A \cap B)}{P(A)}$$

- Both equations describe the same joint probability $P(A \cap B)$

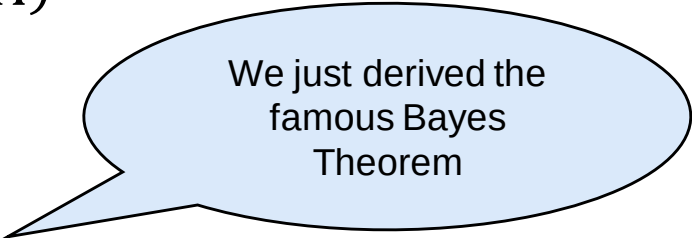
$$P(A \cap B) = P(A|B)P(B) \quad \text{and} \quad P(A \cap B) = P(B|A)P(A)$$

- Equate the two joint forms

$$P(A|B)P(B) = P(B|A)P(A)$$

- Rearrange to express $P(A|B)$

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$



We just derived the
famous Bayes
Theorem

BAYES THEOREM

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Here:

- $P(A|B) \rightarrow$ **Posterior probability:** Probability of event A after observing event B
- $P(B|A) \rightarrow$ **Likelihood:** How likely you are to observe event B if event A is true
- $P(A) \rightarrow$ **Prior probability:** Initial probability of event A before observing any evidence
- $P(B) \rightarrow$ **Evidence:** Total probability of event B is occurring under all possible conditions

EXAMPLE

Compute the probability that an email is spam given that it contains the word 'Lottery'

Email Feature / Class	Spam	Not Spam	Total
Contains 'Lottery'	800	160	960
Does not contain 'Lottery'	1,200	7,840	9,040
Total	2,000	8,000	10,000

- $P(\text{Spam}) = \frac{2,000}{10,000} = 0.2$ (*Prior*)

- $P(\text{Lottery}|\text{Spam}) = \frac{800}{2,000} = 0.4$ (*Likelihood*)

- $P(\text{Lottery}) = \frac{960}{10,000} = 0.096$ (*Evidence*)

$$P(\text{Spam}|\text{Lottery}) = \frac{P(\text{Lottery}|\text{Spam}) \times P(\text{Spam})}{P(\text{Lottery})} = \frac{0.4 \times 0.2}{0.096} = \frac{0.08}{0.096} = 0.833\bar{3}$$

QUESTION

If the likelihood $P(B|A)$ increases while the prior $P(A)$ remains the same, what happens to the posterior $P(A|B)$?

- A. Decreases
- B. Increases ✓
- C. Remains unchanged
- D. Becomes 0

Since $P(A | B) \propto P(B | A) \times P(A)$, **an increase in likelihood increases the posterior**

QUESTION

Compute the probability that an email is spam given that it contains the word 'Offer'

Email Feature / Class	Spam	Not Spam	Total
Contains 'Offer'	600	240	840
Does not contain 'Offer'	1,400	7,760	9,160
Total	2,000	8,000	10,000

- $P(\text{Spam}) = \frac{2,000}{10,000} = 0.2$

- $P(\text{Offer}|\text{Spam}) = \frac{600}{2,000} = 0.3$

- $P(\text{Offer}) = \frac{840}{10,000} = 0.084$

$$P(\text{Spam}|\text{Offer}) = \frac{P(\text{Offer}|\text{Spam}) \times P(\text{Spam})}{P(\text{Offer})} = \frac{0.3 \times 0.2}{0.084} = \frac{0.06}{0.084} = 0.714$$

Bayes Theorem for Multiple Features, Curse of Dimensionality, and Conditional Independence Assumption

BAYES THEOREM FOR MULTIPLE FEATURES

$$P(C|x_1, x_2, \dots, x_n) = \frac{P(x_1, x_2, \dots, x_n|C) P(C)}{P(x_1, x_2, \dots, x_n)}$$

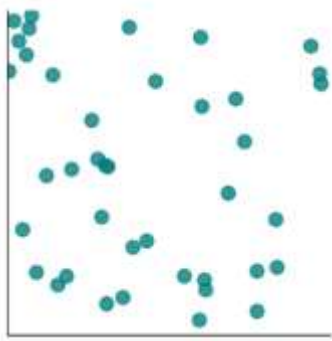
Here:

- $P(C|x_1, x_2, \dots, x_n)$: Posterior probability of class C
- $P(x_1, x_2, \dots, x_n|C)$: Likelihood of seeing all features together
- $P(C)$: Prior probability of the class
- $P(x_1, x_2, \dots, x_n)$: Evidence

CURSE OF DIMENSIONALITY

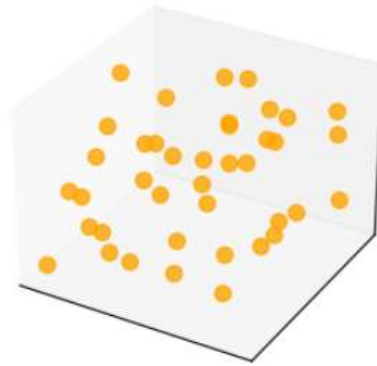
As the number of features increase, data points become sparse in a large feature space

2 Features



Dense region
(Data well covered)

3 Features



Same points → More space
(Sparse coverage)

10 Features



Very sparse
(High-dimensional void)

- $P(x_1, x_2, \dots, x_n|C)$ requires estimating a high-dimensional joint distribution
- With limited data, most combinations of feature values are never observed, making direct estimation unreliable

CONDITIONAL INDEPENDENCE ASSUMPTION

- We cannot estimate the full joint probability $P(x_1, x_2, \dots, x_n|C)$ reliably for many features
- **Assumption:** Once class C is known, the features $x_1, x_2, \dots, x_n$ behave independently

$$P(x_1, x_2, \dots, x_n|C) = \prod_{i=1}^n P(x_i|C)$$

- Each feature contributes its own likelihood $P(x_i|C)$ to the total
- This turns one impossible joint-probability problem into n simple 1-D problems
- The assumption is rarely exact, but it works well when feature dependencies do not change class ranking significantly
- Owing to this simplification, the model is called **Naïve** Bayes

EXAMPLE

House Area (sq. ft)	No. of Rooms	No. of Bathrooms	Location	Price Category (Class)
850.65	2	1	Suburb	Low
1,200.22	3.5	2	Suburb	Low
1,800.50	4	2	City	High
2,200.76	5	3	City	High
1,000.00	2.5	1	City Edge	Low
2,500.32	6	4.5	City Centre	High

- *House Area, Rooms, and Bathrooms* are positively correlated; a larger area → more rooms → more bathrooms
- However, Naïve Bayes assumes they are independent when computing the class probability

Naïve Bayes Classifier

NAÏVE BAYES CLASSIFIER

- Start from the general form

$$P(C|x_1, \dots, x_n) = \frac{P(x_1, \dots, x_n|C)P(C)}{P(x_1, \dots, x_n)}$$

- Apply conditional independence

$$P(C|x_1, \dots, x_n) = \frac{P(C) \prod_i P(x_i|C)}{P(x_1, \dots, x_n)}$$

- Since the denominator $P(x_1, \dots, x_n)$ is constant across classes; for classification, we compare only the numerator

$$P(C|x_1, \dots, x_n) \propto P(C) \prod_i P(x_i|C)$$

DECISION RULE FOR NAÏVE BAYES CLASSIFIER

- In classification, we assign a sample to the class with the highest posterior probability
- Final decision rule

$$\hat{C} = \arg \max_C [P(C) \prod_i P(x_i|C)]$$

↑
Predicted class

This is the **Naïve Bayes classifier**

EXAMPLE

Classify a new email as Spam or Not Spam using two binary features

- x_1 : Contains the word 'Offer'
- x_2 : Contains a link

Class	Total Emails	Emails with word "offer"	Emails with a link
Spam	30	18	21
Not Spam	70	7	14
Total	100	25	35

For Spam

- $P(\text{Spam}) = \frac{30}{100} = 0.3$
- $P(\text{offer} = 1|\text{Spam}) = \frac{18}{30} = 0.6$
- $P(\text{link} = 1|\text{Spam}) = \frac{21}{30} = 0.7$

For Not Spam

- $P(\text{Not Spam}) = \frac{70}{100} = 0.7$
- $P(\text{offer} = 1|\text{Not Spam}) = \frac{7}{70} = 0.1$
- $P(\text{link} = 1|\text{Not Spam}) = \frac{14}{70} = 0.2$

CALCULATING POSTERIOR PROBABILITIES

New email: $X = (\text{offer} = 1, \text{link} = 1)$

$$\text{Posterior}(C) \propto P(C) \prod_i P(x_i = 1|C)$$

$$\begin{aligned}\text{Posterior}(\text{Spam}) &\propto P(\text{Spam}) \times P(\text{offer} = 1|\text{Spam}) \times P(\text{link} = 1|\text{Spam}) \\ &= 0.3 \times 0.60 \times 0.70 = 0.126\end{aligned}$$

$$\begin{aligned}\text{Posterior}(\text{Not Spam}) &\propto P(\text{Not Spam}) \times P(\text{offer} = 1|\text{Not Spam}) \times P(\text{link} = 1|\text{Not Spam}) \\ &= 0.7 \times 0.10 \times 0.20 = 0.014\end{aligned}$$

$\text{Posterior}(\text{Spam}) > \text{Posterior}(\text{Not Spam}) \Rightarrow \boxed{\text{Predicted Class} \rightarrow \text{Spam}}$

Naïve Bayes: Variants

NAÏVE BAYES CLASSIFIER: VARIANTS

- **Multinomial Naïve Bayes:** Used when features represent frequencies or counts, such as the number of times a word appears in an email or document
- **Bernoulli Naïve Bayes:** Used when features are binary, such as whether a word is either *present* or *absent* or a symptom is *experienced* or *not*
- **Gaussian Naïve Bayes:** Used when features are continuous values, such as height, weight, temperature, or any numeric measurement

Multinomial Naïve Bayes

MULTINOMIAL NAÏVE BAYES CLASSIFIER

- Used when features are **discrete counts or frequencies**, such as word occurrences in a document
- It models these counts using the probability mass function (PMF) of the multinomial distribution

$$P(X|C) = \frac{N!}{x_1! x_2! \dots, x_n!} \prod_i P(w_i|C)^{x_i}$$

For classification, constants are dropped

$$P(X|C) \propto \prod_i P(w_i|C)^{x_i}$$

- Posterior probability

$$P(C|X) \propto P(C) \prod_i P(w_i|C)^{x_i}$$

EXAMPLE

Message ID	Message Text	Class
M1	'Buy limited items'	Spam
M2	'Special offer sale'	Spam
M3	'Let's meet tomorrow'	Not Spam
M4	'See you soon'	Not Spam

Determine if the message *'buy items, buy items'* is spam or not spam

EXAMPLE

Vocabulary: {buy, limited, items, special, offer, sale, let's, meet, tomorrow, see, you, soon}

Vocabulary size $V = 12$

Class	Word Counts	Total Words
Spam (M1, M2)	buy: 1, limited: 1, items: 1, special: 1, offer: 1, sale: 1	6
Not Spam (M3, M4)	let's: 1, meet: 1, tomorrow: 1, see: 1, you: 1, soon: 1	6

CALCULATING PROBABILITIES

○ Prior Probabilities

- Spam = $\frac{\text{Spam messages}}{\text{Total messages}} = \frac{2}{2+2} = 0.5$
- Not Spam = $\frac{\text{Not spam messages}}{\text{Total messages}} = \frac{2}{2+2} = 0.5$

○ Estimating word probabilities (using Laplace smoothing)

$$P(w|C) = \frac{\text{count}(w, C) + 1}{\text{total words in } C + V}$$

○ For Spam

- $P(\text{buy}|\text{spam}) = \frac{(1+1)}{(6+12)} = \frac{2}{18}$

- $P(\text{items}|\text{spam}) = \frac{(1+1)}{(6+12)} = \frac{2}{18}$

○ For Not Spam

- $P(\text{buy}|\text{not spam}) = \frac{(0+1)}{(6+12)} = \frac{1}{18}$

- $P(\text{items}|\text{not spam}) = \frac{(0+1)}{(6+12)} = \frac{1}{18}$

CALCULATING POSTERIOR PROBABILITIES

- **Each word appears twice:** $f_{buy} = 2, f_{items} = 2$
- $\text{Posterior}(\text{Spam}|d) \propto P(\text{Spam}) \times (P(\text{buy}|\text{Spam}))^2 \times (P(\text{items}|\text{Spam}))^2$
$$= 0.5 \times \left(\frac{2}{18}\right)^4 = \frac{16}{104976} \approx 1.52 \times 10^{-4}$$
- $\text{Posterior}(\text{Not Spam}|d) \propto P(\text{Not Spam}) \times (P(\text{buy}|\text{Not Spam}))^2 \times (P(\text{items}|\text{Not Spam}))^2$
$$= 0.5 \times \left(\frac{1}{18}\right)^4 = \frac{1}{104976} \approx 9.5 \times 10^{-6}$$

$\text{Posterior}(\text{Spam}|d) > \text{Posterior}(\text{Not Spam}|d) \Rightarrow \boxed{\text{Predicted Class} \rightarrow \text{Spam}}$

Bernoulli Naïve Bayes

BERNOULLI NAÏVE BAYES CLASSIFIER

- Used for **binary features**, where each attribute can be *present* (1) or *absent* (0), and predicts the class based on how likely this combination of yes/no attributes is under each class

- Likelihood of a feature

$$P(X|C) = \prod_i \underbrace{P(x_i|C)^{x_i}}_{\text{Probability that feature } i \text{ appears in class } C} \underbrace{[1 - P(x_i|C)]^{(1-x_i)}}_{\text{Probability that feature } i \text{ does not appear in class } C}$$

- Posterior probability

$$P(C|X) \propto P(C) \prod_i P(x_i|C)^{x_i} [1 - P(x_i|C)]^{(1-x_i)}$$

EXAMPLE

Name	Give Birth	Can Fly	Lives in Water	Have Legs	Class
Human	Yes	No	No	Yes	Mammal
Python	No	No	No	No	Non-Mammal
Salmon	No	No	Yes	No	Non-Mammal
Whale	Yes	No	Yes	No	Mammal
Frog	No	No	Sometimes	Yes	Non-Mammal
Komodo	No	No	No	Yes	Non-Mammal
Bat	Yes	Yes	No	Yes	Mammal
Pigeon	No	Yes	No	Yes	Non-Mammal
Cat	Yes	No	No	Yes	Mammal
Leopard Shark	Yes	No	Yes	No	Non-Mammal
Turtle	No	No	Sometimes	Yes	Non-Mammal
Penguin	No	No	Sometimes	Yes	Non-Mammal
Porcupine	Yes	No	No	Yes	Mammal
Eel	No	No	Yes	No	Non-Mammal
Salamander	No	No	Sometimes	Yes	Non-Mammal
Gila Monster	No	No	No	Yes	Non-Mammal
Platypus	No	No	No	Yes	Mammal
Owl	No	Yes	No	Yes	Non-Mammal
Dolphin	Yes	No	Yes	No	Mammal
Eagle	No	Yes	No	Yes	Non-Mammal

EXAMPLE

Give Birth	Can Fly	Lives in Water	Have Legs	Class
Yes	No	No	Yes	?

EXAMPLE

Name	Give Birth	Can Fly	Lives in Water	Have Legs	Class
Human	Yes	No	No	Yes	Mammal
Python	No	No	No	No	Non-Mammal
Salmon	No	No	Yes	No	Non-Mammal
Whale	Yes	No	Yes	No	Mammal
Frog	No	No	Sometimes	Yes	Non-Mammal
Komodo	No	No	No	Yes	Non-Mammal
Bat	Yes	Yes	No	Yes	Mammal
Pigeon	No	Yes	No	Yes	Non-Mammal
Cat	Yes	No	No	Yes	Mammal
Leopard Shark	Yes	No	Yes	No	Non-Mammal
Turtle	No	No	Sometimes	Yes	Non-Mammal
Penguin	No	No	Sometimes	Yes	Non-Mammal
Porcupine	Yes	No	No	Yes	Mammal
Eel	No	No	Yes	No	Non-Mammal
Salamander	No	No	Sometimes	Yes	Non-Mammal
Gila Monster	No	No	No	Yes	Non-Mammal
Platypus	No	No	No	Yes	Mammal
Owl	No	Yes	No	Yes	Non-Mammal
Dolphin	Yes	No	Yes	No	Mammal
Eagle	No	Yes	No	Yes	Non-Mammal

○ For Mammal (M)

- $P(\text{Give Birth} = 1|M) = \frac{6}{7}$
- $P(\text{Can Fly} = 1|M) = \frac{1}{7}$
- $P(\text{Lives in Water} = 1|M) = \frac{2}{7}$
- $P(\text{Have Legs} = 1|M) = \frac{6}{7}$

○ For Non-Mammal (N)

- $P(\text{Give Birth} = 1|N) = \frac{1}{13}$
- $P(\text{Can Fly} = 1|N) = \frac{3}{13}$
- $P(\text{Lives in Water} = 1|N) = \frac{9}{13}$
- $P(\text{Have Legs} = 1|N) = \frac{9}{13}$

EXAMPLE

Give Birth	Can Fly	Lives in Water	Have Legs	Class
Yes	No	No	Yes	?

○ Give Birth = 1, Can Fly = 0, Lives in Water = 0, Have Legs = 1

○ For Mammal (M)

$$\begin{aligned}P(A|M) &= P(GB = 1|M)[1 - P(CF = 1|M)][1 - P(LW = 1|M)]P(HL = 1|M) \\&= \frac{6}{7} \times \left(1 - \frac{1}{7}\right) \times \left(1 - \frac{2}{7}\right) \times \frac{6}{7} = \frac{6}{7} \times \frac{6}{7} \times \frac{5}{7} \times \frac{6}{7} \approx 0.06\end{aligned}$$

○ For Non-Mammal (N)

$$\begin{aligned}P(A|N) &= P(GB = 1|N)[1 - P(CF = 1|N)][1 - P(LW = 1|N)]P(HL = 1|N) \\&= \frac{1}{13} \times \left(1 - \frac{3}{13}\right) \times \left(1 - \frac{9}{13}\right) \times \frac{9}{13} = \frac{1}{13} \times \frac{10}{13} \times \frac{4}{13} \times \frac{9}{13} \approx 0.0042\end{aligned}$$

EXAMPLE

Give Birth	Can Fly	Lives in Water	Have Legs	Class
Yes	No	No	Yes	Mammal

- Multiply by class priors

$$P(M) = \frac{7}{20} = 0.35 \quad P(N) = \frac{13}{20} = 0.65$$

$$\text{Posterior}(M) \propto P(A|M) \times P(M) = 0.06 \times 0.35 = 0.021$$

$$\text{Posterior}(N) \propto P(A|N) \times P(N) = 0.0042 \times 0.65 = 0.0027$$

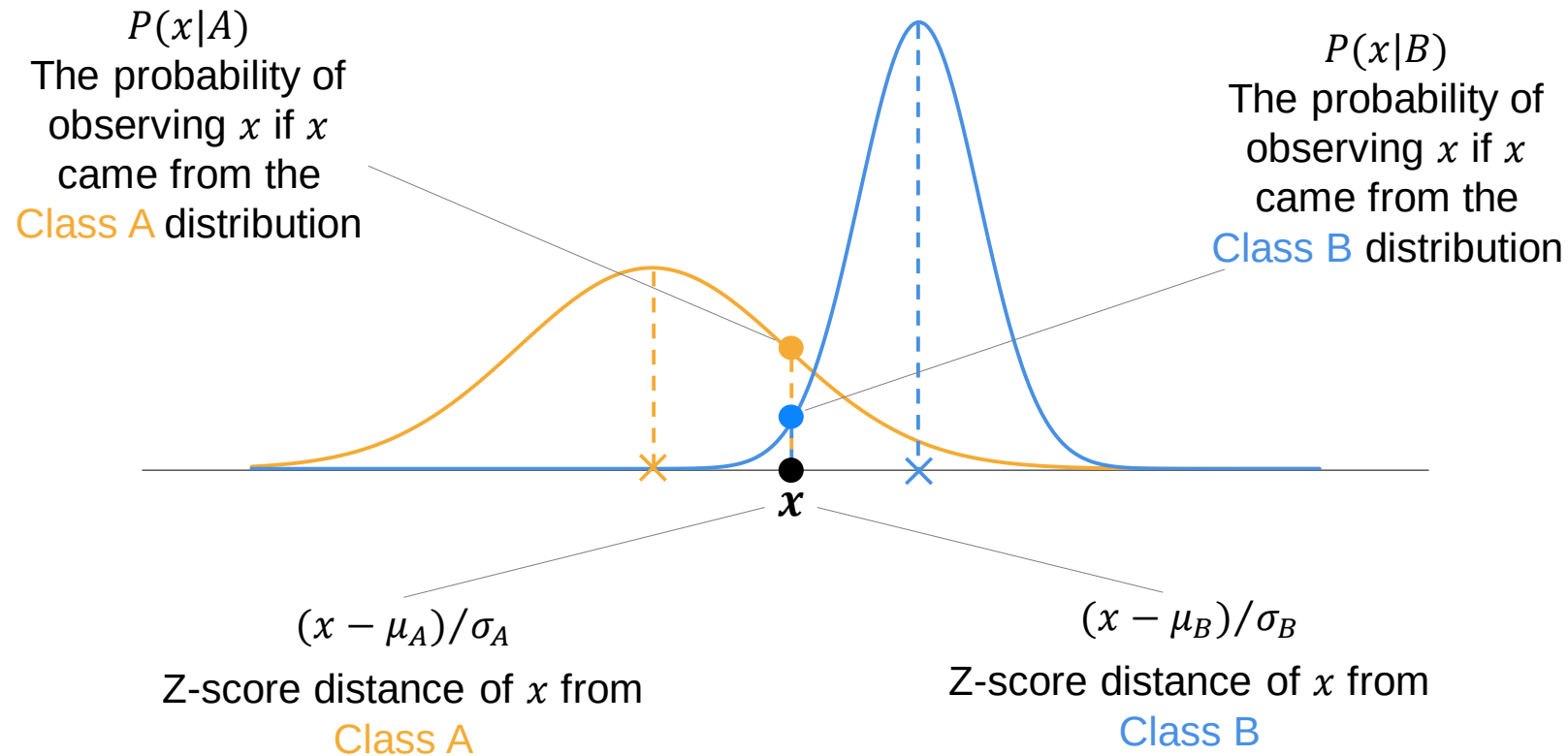
- Compare posteriors

$$\text{Posterior}(M) > \text{Posterior}(N) \Rightarrow \boxed{\text{Predicted Class} \rightarrow \text{Mammal}}$$

Gaussian Naïve Bayes

GAUSSIAN NAÏVE BAYES CLASSIFIER

Used when features are **continuous**, such as height, weight, temperature or sensor readings and assumes that the values for each feature in a class follow a normal (Gaussian) distribution



Gaussian Naïve Bayes chooses the class whose distribution gives the higher probability density for x

GAUSSIAN NAÏVE BAYES CLASSIFIER

- **Likelihood of a feature:** We use the probability density function (PDF) of the normal distribution to estimate how likely a continuous feature value x_i is for a given class C

$$P(x_i|C) = \frac{1}{\sqrt{2\pi\sigma_C^2}} e^{-\frac{(x_i-\mu_C)^2}{2\sigma_C^2}}$$

- Posterior probability

$$P(C|X) \propto P(C) \prod_i \frac{1}{\sqrt{2\pi\sigma_{C_i}^2}} e^{-\frac{(x_i-\mu_{C_i})^2}{2\sigma_{C_i}^2}}$$

Each feature's Gaussian likelihood is multiplied with the class prior $P(C)$, and the class with the higher posterior becomes the prediction

EXAMPLE

Height (Feet)	Weight (Kg)	Foot Size (Inch)	Class
6	84	10	Male
6.2	88	11	Male
5.76	74	9	Male
4.11	48	5	Female
5.2	55	6	Female
5.92	71	10	Male
5.5	57	8	Female
5	52	7	Female

Height (Feet)	Weight (Kg)	Foot Size (Inch)	Class
5.6	59	7.5	?

- Posterior (M) $\propto P(M) \times P(H|M) \times P(W|M) \times P(FS|M)$
- Posterior (F) $\propto P(F) \times P(H|F) \times P(W|F) \times P(FS|F)$

CALCULATING PROBABILITIES

Height (Feet)	Weight (Kg)	Foot Size (Inch)	Class
6	84	10	Male
6.2	88	11	Male
5.76	74	9	Male
4.11	48	5	Female
5.2	55	6	Female
5.92	71	10	Male
5.5	57	8	Female
5	52	7	Female

Prior Probabilities

- $$P(M) = \frac{\text{Number of male samples}}{\text{Number of male + Female samples}}$$
$$= \frac{4}{4 + 4} = 0.5$$

- $$P(F) = \frac{\text{Number of female samples}}{\text{Number of male + Female samples}}$$
$$= \frac{4}{4 + 4} = 0.5$$

MALE CLASS: MEAN AND VARIANCE CALCULATIONS

○ Height

- Mean = $\frac{6 + 6.2 + 5.76 + 5.92}{4} = \frac{23.88}{4} = 5.97$

- Variance = $(SD)^2 = \sigma^2 = \sum \frac{(x - \bar{x})^2}{n - 1}$

$$\begin{aligned}\sigma^2 &= \frac{(6 - 5.97)^2 + (6.2 - 5.97)^2 + (5.76 - 5.97)^2 + (5.92 - 5.97)^2}{4 - 1} \\ &= \frac{0.0009 + 0.0529 + 0.0441 + 0.0025}{3} = 0.033\end{aligned}$$

MALE CLASS: MEAN AND VARIANCE CALCULATIONS

○ Weight

- Mean = $\frac{84 + 88 + 74 + 71}{4} = 79.25$

- Variance = $\sigma^2 = \frac{(84 - 79.25)^2 + (88 - 79.25)^2 + (74 - 79.25)^2 + (71 - 79.25)^2}{4 - 1}$

$$\sigma^2 = \frac{22.56 + 76.56 + 27.56 + 68.06}{3} = 64.91$$

○ Foot size

- Mean = $\frac{10 + 11 + 9 + 10}{4} = 10$

- Variance = $\sigma^2 = \frac{(10 - 10)^2 + (11 - 10)^2 + (9 - 10)^2 + (10 - 10)^2}{4 - 1}$

$$\sigma^2 = \frac{0 + 1 + 1 + 0}{3} = 0.66$$

FEMALE CLASS: MEAN AND VARIANCE CALCULATIONS

○ Height

- Mean = 4.95
- Variance = $\sigma^2 = 0.354$

○ Weight

- Mean = 53
- Variance = $\sigma^2 = 18$

○ Foot size

- Mean = 6.5
- Variance = $\sigma^2 = 1.66$

CALCULATING POSTERIOR PROBABILITIES

Male: Mean and Variance Values

Feature	Height	Weight	Foot Size
Mean	5.97	79.25	10
Variance	0.033	64.91	0.66

Female: Mean and Variance Values

Feature	Height	Weight	Foot Size
Mean	4.95	53	6.5
Variance	0.354	18	1.66

Height (Feet)	Weight (Kg)	Foot Size (Inch)	Class
5.6	59	7.5	?

- $\text{Posterior}(M) \propto P(M) \times P(H|M) \times P(W|M) \times P(FS|M)$
- $\text{Posterior}(F) \propto P(F) \times P(H|F) \times P(W|F) \times P(FS|F)$

CALCULATING THE POSTERIOR PROBABILITY: MALE

Height	Weight	Foot Size	Class
5.6	59	7.5	?

- Probability of Height being 5.6 given that person is Male

$$P(H|M) = \frac{1}{\sqrt{2 \times 3.14 \times 0.033}} e^{-\frac{(5.6-5.97)^2}{2 \times 0.033}} = \frac{1}{0.455} e^{-\frac{0.1369}{0.066}} = 2.197 \times 0.1256 = 0.276$$

- Probability of Weight being 59 given that person is Male

$$P(W|M) = \frac{1}{\sqrt{2 \times 3.14 \times 64.91}} e^{-\frac{(59-79.25)^2}{2 \times 64.91}} = \frac{1}{20.18} e^{-\frac{410.06}{129.82}} = 0.049 \times 0.0425 = 0.0020$$

- Probability of Foot Size being 7.5 given that person is Male

$$P(FS|M) = \frac{1}{\sqrt{2 \times 3.14 \times 0.66}} e^{-\frac{(7.5-10)^2}{2 \times 0.66}} = \frac{1}{2.035} e^{-\frac{6.25}{1.32}} = 0.491 \times 0.0087 = 0.00431$$

CALCULATING THE POSTERIOR PROBABILITY: MALE

- $P(M) = 0.5$
- $P(H|M) = 0.276$
- $P(W|M) = 0.0020$
- $P(FS|M) = 0.00431$

$$\text{Posterior (M)} \propto P(M) \times P(H|M) \times P(W|M) \times P(FS|M)$$

$$= 0.5 \times 0.276 \times 0.0020 \times 0.0043 = \mathbf{0.0000011}$$

CALCULATING THE POSTERIOR PROBABILITY: FEMALE

Height	Weight	Foot Size	Class
5.6	59	7.5	?

- Probability of Height being 5.6 given that person is Female

$$P(H|F) = \frac{1}{\sqrt{2 \times 3.14 \times 0.354}} e^{-\frac{(5.6-4.95)^2}{2 \times 0.354}} = \frac{1}{2.22} e^{-\frac{0.3035}{0.708}} = 0.45 \times 0.6514 = 0.293$$

- Probability of Weight being 59 given that person is Female

$$P(W|F) = \frac{1}{\sqrt{2 \times 3.14 \times 18}} e^{-\frac{(59-53)^2}{2 \times 18}} = \frac{1}{10.63} e^{-\frac{36}{36}} = 0.094 \times 0.3678 = 0.0345$$

- Probability of Foot Size being 7.5 given that person is Female

$$P(FS|F) = \frac{1}{\sqrt{2 \times 3.14 \times 1.66}} e^{-\frac{(7.5-6.5)^2}{2 \times 1.66}} = \frac{1}{3.228} e^{-\frac{1}{3.228}} = 0.309 \times 0.3012 = 0.093$$

CALCULATING POSTERIOR PROBABILITY: FEMALE

- $P(F) = 0.5$
- $P(H|F) = 0.293$
- $P(W|F) = 0.0345$
- $P(FS|F) = 0.093$

$$\text{Posterior (F)} \propto P(F) \times P(H|F) \times P(W|F) \times P(FS|F)$$

$$= 0.5 \times 0.293 \times 0.0345 \times 0.093 = \mathbf{0.00047}$$

COMPARING POSTERIOR PROBABILITIES

- $\text{Posterior}(\text{M}) = 0.0000011$

- $\text{Posterior}(\text{F}) = 0.00047$

$$\text{Posterior}(\text{M}) < \text{Posterior}(\text{F}) \Rightarrow \boxed{\text{Predicted Class} \rightarrow \text{Female}}$$

Underflow Issues & Log Probabilities

NUMERICAL UNDERFLOW

- Naïve Bayes multiplies many small probabilities

$$P(C|X) \propto P(C) \prod_i P(x_i|C)$$

- Each term $P(x_i|C) < 1 \rightarrow$ Product rapidly shrinks

$$P(C|X) = 0.5 \times 0.002 \times 0.01 \times \dots \times 0.003 \approx 0.0$$

- Extremely small values $\rightarrow$ **Numerical underflow**
- Once this happens, all class scores may look identical

EXAMPLE

Feature	$P(x_i C)$
Word 1	0.07
Word 2	0.04
Word 3	0.02
⋮	⋮
Word 248	0.01
Word 249	0.04
Word 250	0.02

$$\begin{aligned} P(C|X) &= P(C) \prod_i P(x_i|C) \\ &= 0.5 \times 0.07 \times 0.04 \times 0.02 \times \cdots \times 0.01 \times 0.04 \times 0.02 \approx 1.1 \times 10^{-420} \end{aligned}$$

Underflows to 0 in standard floating-point arithmetic

LOG PROBABILITIES

- Instead of multiplying probabilities, we can take their logarithms

$$\log P(C|X) = \log P(C) + \sum_i \log P(x_i|C)$$

- Logs avoids shrinkage and keep computation stable
- Since logarithms are monotonic, the class with the higher log-posterior still corresponds to the higher true probability

EXAMPLE

Feature	$P(x_i C)$	$\log P(x_i C)$
Word 1	0.07	-2.66
Word 2	0.04	-3.22
Word 3	0.02	-3.91
.	.	.
.	.	.
.	.	.
Word 248	0.01	-4.61
Word 249	0.04	-3.22
Word 250	0.02	-3.91

$$\log P(C|X) = \log P(C) + \sum_i \log P(x_i|C)$$

$$= (-0.693) + (-2.66) + (-3.22) + (-3.91) + \dots + (-4.61) + (-3.22) + (-3.91) = -967.2$$

Log-space keeps the value representable and avoids underflow

Advantages and Disadvantages of Naive Bayes

ADVANTAGES AND DISADVANTAGES OF NAÏVE BAYES CLASSIFIER

ADVANTAGES

- Fast for training and inference
- Works well with high dimensional data
- Simple and interpretable
- Supports multiple data types

DISADVANTAGES

- Strong feature independence assumption
- Cannot model feature interactions
- Gaussian assumption for continuous data